

COSC341 - Assignment 3 - Proving Dominating Set is $\mathcal{NP}$ -Complete

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1 The Dominating Set Problem

In graph theory, a graph G is defined by a vertex set V and an edge set E which contains unordered pairs of distinct elements of V . Nodes v and w are adjacent if there is an edge $\{v, w\} \in E$. A dominating set for a graph $G = (V, E)$ is a subset $D \subseteq V$ of its vertices, such that any vertex in V is either in D , or is adjacent to a vertex in D .

The Dominating Set problem is a decision problem to decide whether or not there is a dominating set of a specific size for a given graph. An example dominating set of size 2 for the given graph is shown in Figure 1 where the vertices coloured in red are the vertices in this particular dominating set.

DOMINATING-SET

Instance: A graph G and a positive integer k .

Problem: Is there a dominating set in G having k elements?

To show that this problem is $\mathcal{NP}$ -Complete we need to show that it is both in $\mathcal{NP}$ and that it is $\mathcal{NP}$ -hard as we can reduce to it in polynomial time from another problem that is $\mathcal{NP}$ -Complete.

2 Showing Dominating Set is in $\mathcal{NP}$

In order to show that the Dominating Set problem is in $\mathcal{NP}$ we need to show that if we are given a solution, that it can be verified in polynomial time. Suppose an oracle gives us a set $D \subseteq V$. The goal is to check if this set D is a dominating set of G in polynomial time.

For G , we can verify a possible dominating set (D) by checking, for each vertex in V , whether at least one of v 's neighbours is in D using the below algorithm.

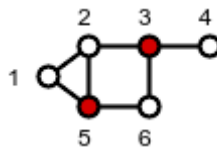


Figure 1: An Example Dominating Set from [1]

Algorithm

```
dominating_set = {...}

for v in vertices
    dominated = false
    if v in dominating_set
        dominated = true
    else
        for n in v.neighbours
            if n in dominating_set
                dominated = true
                break

    if dominated == false
        return false

return true
```

Complexity of The Algorithm

For each vertex $v \in V$ we check if v is in the dominating set D which is constant work. Since each vertex is processed once, we have $O(|V|)$ for the outer loop. For the inner loop we run through all of the edge pairs. By the *Handshake Lemma*¹, which states that sum of all degrees is equal to twice the edges in a graph, we get $O(|E|)$ for the inner loop. The total complexity of the algorithm is $O(|V| + |E|)$, which is linear, or more broadly the algorithm has polynomial runtime. Therefore, the Dominating Set problem is in $\mathcal{NP}$.

3 Reducing from a known $\mathcal{NP}$ -Complete problem

We now need to show that we can reduce the Dominating Set problem from a known NP-Complete problem in polynomial time. As we have worked through in lectures the Vertex Cover problem is an NP-Complete problem so we will demonstrate reducing the Dominating Set problem from the Vertex Cover problem and show that it can be done in polynomial time.

VERTEX-COVER

Instance: A graph G and a positive integer k

Problem: Is there a subset of the vertices of G having at most k elements which contains at least one of the endpoints of every edge?

We need an algorithm to take a graph $G = (V, E)$ and give a new graph $G' = (V', E')$ that has a dominating set of size k , if and only if vertex cover of size k exists in G . We start with initialising the vertex set V' for G' by duplicating the vertex set V from G . Then we process the edges in E of G .

¹For details, see [2]

For every edge $\{u, v\}$ in the original G we create a triangle of the original vertices u and v and a new vertex uv in G' by adding a new vertex uv to V' and 3 edges $\{u, v\}, \{uv, u\}, \{uv, v\}$ to E' . This can obviously be done in polynomial time in the number of vertices and edges in the original graph G and we will show that this is all that is required.

If the original graph G had a vertex cover of size k then every edge pair $\{u, v\}$ in G has at least one of u or v in the vertex cover set. Without loss of generality, if u is in the vertex cover of G , then it is adjacent to both v and the new uv vertex in G' . So we can add u to the dominating set for G' . In this way the set of k vertices forming the vertex cover of G form the dominating set of size k for G' .

On the other hand, if the graph G' has a dominating set of size k then for each vertex in the dominating set, it is one of the original vertices from G or it is one of the additional vertices added. In this second case, as each new vertex is part of a triangle (where the other two vertices are from the original graph G), we can replace it in the dominating set with one of the other vertices of the triangle without changing the span or size of the dominating set. In this way the dominating set can be modified to remove all references to the additional vertices of G' while still having a dominating set that spans them, although this set may then contain duplicate vertices. This set is also then a vertex cover for the original graph G with at least either the u or v vertex being in the set for each edge $\{u, v\}$. So, if G' has a dominating set of size k , then G will have a vertex cover with size at most k .

Therefore, we can say that a graph G' contains a dominating set of size k if and only if the graph G contains a vertex cover of at most size k . We have also shown an algorithm that will reduce an instance of the vertex cover problem to an instance of the dominating set problem in polynomial time. Combining this with the knowledge that the dominating set problem is in $\mathcal{NP}$ and the vertex cover problem is $\mathcal{NP}$ -Complete, we conclude that the dominating set problem is also $\mathcal{NP}$ -Complete.²

References

- [1] “Dominating set,” May 2026, page Version ID: 1352191481. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Dominating_set&oldid=1352191481
- [2] “Handshaking lemma,” Dec. 2025, page Version ID: 1329677553. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Handshaking_lemma&oldid=1329677553
- [3] “Proof that Dominant Set of a Graph is NP-Complete,” Jul. 2025. [Online]. Available: <https://www.geeksforgeeks.org/dsa/proof-that-dominant-set-of-a-graph-is-np-complete/>

²This particular argument was found here: [3]